

Tutorial Handout: Absolute Discounting

1. Intuition: Why a Constant Discount D ?

Before Absolute Discounting, algorithms like Jelinek-Mercer used **proportional discounting** ($\lambda \cdot p_{\text{ML}}$), which scaled down large counts by huge numbers and small counts by tiny numbers [224–225].

Church & Gale (1991) discovered empirically that in real language corpora, the average number of counts you need to "subtract" from observed n -grams to make room for unseen ones is **roughly flat/constant** regardless of whether the n -gram occurred 3 times or 30 times [73, 222–223].

The Core Rule of Absolute Discounting:

Subtract a **fixed, constant discount** D (e.g., $D = 0.5$ or 0.75) from every observed count.

2. The Formula

For a bigram model [71–72]:

$$p_{\text{abs}}(w_i \mid w_{i-1}) = \frac{\max(c(w_{i-1}w_i) - D, 0)}{c(w_{i-1})} + \left(\frac{D}{c(w_{i-1})} \cdot N_{1+}(w_{i-1} \cdot) \right) p_{\text{lower}}(w_i)$$

- **First Term (Discounted High-Order):** Subtracts D from every observed bigram count.
- **Second Term (Interpolation Weight):** Collects all the subtracted mass into a pool and hands it down to the lower-order distribution [71–72].
- $N_{1+}(w_{i-1} \cdot)$: The number of unique word types that have appeared following history w_{i-1} .

3. Step-by-Step Toy Example

Setup:

- Training Corpus: "cat sat", "cat slept", "dog sat"
- Vocabulary $V = \{ \text{"cat", "dog", "sat", "slept"} \}$
- History $w_{i-1} = \text{"cat"}$ ($c(\text{"cat"}) = 2$) **Target: "cat sat" ; "cat dog"**

- Observed bigrams: $c(\text{"cat sat"}) = 1, c(\text{"cat slept"}) = 1$
- Set discount $D = 0.5$

Step A: Calculate Lower-Order Probabilities (Raw Unigrams)

Total words $N = 6$.

- $p_{\text{ML}}(\text{"sat"}) = \frac{2}{6} = 0.333$
- $p_{\text{ML}}(\text{"slept"}) = \frac{1}{6} = 0.167$
- $p_{\text{ML}}(\text{"dog"}) = \frac{1}{6} = 0.167$

Step B: Compute Bigram Probabilities

1. For Seen Bigram "cat sat" :

$$\text{Discounted term} = \frac{1 - 0.5}{2} = \frac{0.5}{2} = 0.25$$

$$\text{Leftover Weight } \lambda(\text{"cat"}) = \frac{0.5}{2} \times 2 \text{ unique words after 'cat'} = 0.50$$

$$p_{\text{abs}}(\text{"sat"} \mid \text{"cat"}) = 0.25 + (0.50 \times 0.333) = 0.25 + 0.166 = \mathbf{0.416}$$

2. For Unseen Bigram "cat dog" :

$$\text{Discounted term} = \frac{\max(0 - 0.5, 0)}{2} = 0.00$$

$$p_{\text{abs}}(\text{"dog"} \mid \text{"cat"}) = 0.00 + (0.50 \times 0.167) = \mathbf{0.0835}$$

4. What Makes Absolute Discounting Better than Earlier Methods?

- **Better than Add-1 / Add- δ :** It doesn't invent fake counts for every vocabulary word, which severely distorts probabilities [39–40].
- **Better than Jelinek-Mercer:** Fixed subtraction D matches real corpus statistics much better than linear percentage scaling [223–225].

5. Its Flaw (Why Kneser-Ney comes next)

Notice in Step A that $p_{\text{lower}}(w_i)$ uses **raw unigram frequencies**. This leads directly to the **"San Francisco" problem**: if a word is frequent overall, Absolute Discounting assigns it high

probability in unseen contexts, even if that word only ever appears after one specific history [75–76].